MAT 528: Foundations of Topology

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Definition 1.1 (Topological space) — Given a set X, a topology $\mathcal{T}$ on X is a collection of subsets of X, called **open sets**, such that

(1) $\varnothing$ and X are in $\mathcal{T}$.

(2) If $\{U_\alpha\}_{\alpha\in I}$ is an arbitrary subcollection of $\mathcal{T}$, then $\bigcup_{\alpha\in I} U_\alpha$ is also in $\mathcal{T}$.

(3) If $U_1, \ldots, U_n$ are finitely many sets in $\mathcal{T}$, then $U_1 \cap \cdots \cap U_n$ is in $\mathcal{T}$ as well.

In English, (2) says that arbitrary unions of open sets are also open sets and (3) says that finite intersections of open sets are also open sets. The pair $(X, \mathcal{T})$ is a **topological space**, which, if the underlying topology is clear from context, will just be referred to as X.

Example 1.2

Suppose $X = \{a, b, c\}$ is a 3-element set. Here are some topologies on X:

(a) $\mathcal{T} = \{\varnothing, X\}$; in general, this is called the **trivial topology**.

(b) $\mathcal{T} = \mathcal{P}(X)$, the power set of X; in general, this is called the **discrete topology**.

(c) $\mathcal{T} = \{\{a\}, \varnothing, X\}$.

(d) $\mathcal{T} = \{\{a\}, \{a, b\}, \varnothing, X\}$.

Some non-examples: $\{\{a\}, \{b\}, \varnothing, X\}$ and $\{\{a, b\}, \{a, c\}, \varnothing, X\}$ aren't topologies on X.

A very standard example that lends itself well to visual intuition is the following:

Example 1.3 (Euclidean topology on $\mathbb{R}$)

Give $\mathbb{R}$ a topology by defining $U \subseteq \mathbb{R}$ to be open if for all $x \in U$, there is $\varepsilon > 0$ such that $(x - \varepsilon, x + \varepsilon) \subseteq \mathbb{R}$. Checking that this is a topology:

(1) $\varnothing$ and X are trivially open by this definition (check it!).

(2) If $\{U_\alpha\}_{\alpha\in I}$ is an arbitrary collection of open subsets of $\mathbb{R}$, then any $x \in \bigcup_{\alpha\in I} U_\alpha$ is contained in one of the U_αs, say U_β; then, there exists an $\varepsilon > 0$ such that $(x - \varepsilon, x + \varepsilon) \subseteq U_\beta \subseteq \bigcup_{\alpha\in I} U_\alpha$ so the union of the U_α is open.

(3) If $U_1, \ldots, U_n$ are open and x lies in their intersection, then it of course lies in each U_i. Thus for each $i = 1, \ldots, n$, there is $\varepsilon_i > 0$ such that $(x - \varepsilon_i, x + \varepsilon_i) \subseteq U_i$. Therefore, letting ε be the positive number $\min_{i=1,\ldots,n} \varepsilon_i$, we see that $U_1 \cap \cdots \cap U_n$ is open because $(x - \varepsilon, x + \varepsilon) \subseteq (x - \varepsilon_i, x + \varepsilon_i) \subseteq U_i$ for each i.

Exercise 1.4. Extrapolate the construction of Example 1.3 to get a topology on $\mathbb{R}^n$ (there are a few natural ways to do this; try to show that they all result in the same topology).

Definition 1.5 (Basis) — Let $\mathcal{T}$ be a topology of X. Then, $\mathcal{B} \subseteq \mathcal{T}$ is a **basis** for $\mathcal{T}$ if any $U \subseteq \mathcal{T}$ can be written as the union of some elements of $\mathcal{B}$.

Example 1.6 (Basis of the Euclidean topology)

Consider $\mathbb{R}$ with the Euclidean topology. Then, $\mathcal{B} = \{(a, b) : a, b \in \mathbb{R},\ a < b\}$ is a basis because if U is an open subset of $\mathbb{R}$, then for all $x \in U$, there is an $\varepsilon_x > 0$ such that $(x - \varepsilon_x, x + \varepsilon_x) \subseteq U$, which implies that

$$\bigcup_{x \in U} (x - \varepsilon_x, x + \varepsilon_x) = U.$$

The scenario in the preceding example happens more generally, as we can take note of the fact that the open sets in U were essentially defined with respect to the basis of open intervals.

Theorem 1.7 (Egg yolk property)**.** Let $\mathcal{T}$ be a topology on X and let $\mathcal{B}$ be some collection of sets in $\mathcal{T}$. Then, $\mathcal{B}$ is a basis for $\mathcal{T}$ if and only if for all open sets U and all $x \in U$, there exists $B \in \mathcal{B}$ such that $x \in B \subseteq U$.

Proof. If $\mathcal{B}$ is a basis, then any open U can be written as $\bigcup_{\alpha \in I} B_\alpha$ for some subcollection $\{B_\alpha\}_{\alpha \in I}$ of $\mathcal{B}$. Then, any $x \in U$ is contained in B_β for some $\beta \in I$, so $x \in B_\beta \subseteq U$.

Conversely, if $\mathcal{B}$ has the given property, then given open U, associate with each $x \in U$ a set $B_x \in \mathcal{B}$ such that $x \in B_x \subseteq U$. Then,

$$\bigcup_{x \in U} B_x = U$$

so U is a union of sets in $\mathcal{B}$, meaning $\mathcal{B}$ is a basis for the given topology on X. $\square$

Remark 1.7.1. We called this the "egg yolk property" in class because the picture of $x \in U \subseteq X$, when drawing U and X as blobs, looks like an egg X with a yolk U. This property is really a more general principle: if for each $x \in U$ one can find an open set V (often a basis element) such that $x \in V \subseteq U$, then U must be an open set as well.

The previous theorem can be interpreted as saying that given basis $\mathcal{B}$ for topology $\mathcal{T}$ on X, the sentence "for all $x \in U$, there exists $B \in \mathcal{B}$ such that $x \in B \subseteq U$" means the same thing as "U is open". Observing this lets us quickly prove a nice criterion for checking whether two topologies of X are equal just by looking at a basis of each topology.

Theorem 1.8 (Bases give same topology)**.** Let $\mathcal{B}_1$ be a basis for topology $\mathcal{T}_1$ on X, and let $\mathcal{B}_2$ be a basis for topology $\mathcal{T}_2$ on X. Then, $\mathcal{T}_1 = \mathcal{T}_2$ if and only if for all $x \in X$, given $B_1 \in \mathcal{B}_1$ containing x, there is a $B_2 \in \mathcal{B}_2$ such that $x \in B_2 \subseteq B_1$, and vice versa.

Proof. Observe that the given condition is equivalent to saying that each B_1 is open with respect to $\mathcal{T}_2$ and vice versa; therefore, if U is open in $\mathcal{T}_1$, it is a union of sets in $\mathcal{B}_1$, hence also a union of sets that are open in $\mathcal{T}_2$, hence also open with respect to $\mathcal{T}_2$; and vice versa. That is, $\mathcal{T}_1$ and $\mathcal{T}_2$ have the same collection of open sets! $\square$

Thus, the topologies on $\mathbb{R}^2$ generated by (1) the rectangles $(a,b)\times(c,d)$ and (2) the ε-balls $B_\varepsilon(x)=\{y\in\mathbb{R}^2 : |x-y|<\varepsilon\}$ are easily shown to be the same.

It is also sometimes helpful to consider a more minimal way of generating a topology:

Definition 1.9 (Subbasis) — Let S be a collection of subsets of X whose union is X. Then, the **topology generated by** S, denoted $\mathcal{T}(S)$, consists of arbitrary unions of finite intersections of sets in S. S is called a **subbasis** of $\mathcal{T}(S)$.

Exercise 1.10. Check that $\mathcal{T}(S)$ is indeed a topology (thus the set of finite intersections of elements of S is a basis of $\mathcal{T}(S)$). Also, convince yourself that a basis for a topology is also a subbasis for the same topology.

It is helpful to think of the generated topology as the "smallest" topology on X that contains S, in the following sense:

Definition 1.11 (Fine/coarse) — Let $\mathcal{T}_1$ and $\mathcal{T}_2$ be two topologies on a set X. We say that $\mathcal{T}_1$ is **finer** (or larger) than $\mathcal{T}_2$ and $\mathcal{T}_2$ is **coarser** (or smaller) than $\mathcal{T}_1$ if $\mathcal{T}_1\supseteq\mathcal{T}_1$.

That is:

Theorem 1.12 (Generated topology is minimal). Let S be a subbasis of the topology $\mathcal{T}$. Then, $\mathcal{T}$ is the unique coarsest topology containing S, meaning if $\mathcal{T}'$ contains S as well, then $\mathcal{T}\subseteq\mathcal{T}'$.

Proof. Clearly if $\mathcal{T}'$ contains S then it contains all finite intersections of elements of S and arbitrary unions of these finite intersections, so $\mathcal{T}\subseteq\mathcal{T}'$.

Furthermore, if $\mathcal{T}$ and $\mathcal{T}'$ both contain S and are as coarse as possible, then by definition of "maximally coarse" we have both $\mathcal{T}\subseteq\mathcal{T}'$ and $\mathcal{T}'\subseteq\mathcal{T}$. □

Exercise 1.13. Let S be a subbasis for the topology $\mathcal{T}(S)$ on X. Show that $\mathcal{T}(S)$ is the intersection of all the topologies on X containing S (you may want to first convince yourself that, in general, the intersection of two topologies on X is also a topology on X).

Exercise 1.14. Show that the set of open rays, in $\mathbb{R}$, i.e.

$$\{(a,\infty) : a\in\mathbb{R}\}\cup\{(\infty,b) : b\in\mathbb{R}\},$$

is a subbasis for the Euclidean topology on $\mathbb{R}$.

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Definition 2.1 (Product topology) — Let X and Y be topological spaces. The **product topology** on $X\times Y$ is given by the basis

$$\{U\times V : U\subseteq X \text{ and } V\subseteq Y \text{ are open}\}.$$

(In other words, open sets in $X \times Y$ look like arbitrary unions of boxes.)

Exercise 2.2. Verify that this basis really is a basis for $X \times Y$ (as opposed to a subbasis; you'll need to check that the set is closed under finite intersections, see Definition 1.9).

Lemma 2.3. If $\mathcal{B}_X$ and $\mathcal{B}_Y$ are bases for topologies on X and Y, respectively, then the product topology has basis $\mathcal{B} = \{U \times V : U \in \mathcal{B}_X, V \in \mathcal{B}_Y\}$.

Proof. Let (x, y) be a point in $X \times Y$ and U an open set in $X \times Y$. Then, by definition of product topology, there exists open $U_X \subseteq X$ and $U_Y \subseteq Y$ such that $(x, y) \in U_X \times U_Y \subseteq U$. Pick a basis element $B_X \in \mathcal{B}_X$ so that $x \in B_X \subseteq U_X$, and similarly pick $B_Y \in \mathcal{B}_Y$ so $y \in B_Y \subseteq U_Y$. Then we get

$$(x, y) \in B_X \times B_Y \subseteq U_X \times U_Y \subseteq U$$

so the egg yolk property is satisfied and $\mathcal{B}$ is a basis. $\square$

Thus, one can impose a topology on $\mathbb{R}^2$ by giving open rectangles as a basis. Of course, this is equal to the topology given by ε-balls, as shown in the case of $n = 2$ last lecture.

Definition 2.4 (Subspace topology) — Let Y be a subset of topological space X. Then, Y can be made a topological space by declaring a subset U of Y to be open if and only if $U = V \cap Y$ for some open subset V of X. This is called the **subspace topology** or **topology induced by** X or other phrases of the same ilk.

Example 2.5

Give $X = [0, 1) \cup \{2\}$ the subspace topology it inherits as a subset of $\mathbb{R}$. Then, $\{2\}$ and $[0, 1)$ are open subsets of X, even though they aren't open in $\mathbb{R}$ (check the egg yolk property at 0 and 2 to show this).

Exercise 2.6. Given a basis $\mathcal{B}_X$ for X and $Y \subseteq X$, what is the obvious basis $\mathcal{B}_X$ for the topology of Y? Prove that this really is a basis.

Definition 2.7 (Closed set) — A subset $E \subseteq X$ is closed if it is equal to U^c, the complement $X \setminus U$, of an open subset U.

Example 2.8

The set $[0, 1]$ is closed in $\mathbb{R}$ because it is the complement of the union of open sets $(-\infty, 0) \cup (1, \infty)$, while $[0, 1)$ is neither open nor closed — it isn't open because egg yolk property fails at 0, and it isn't closed because its complement fails egg yolk property at 1.

Proposition 2.9. Closed sets satisfy the following union and intersection properties:

(1) $\varnothing$ and X are closed.

(2) Arbitrary intersections of closed sets are closed.

(3) Finite unions of closed sets are closed.

In principle, one could use closed sets to define a topology instead of open sets; the convention just happens to be open sets for most purposes (although for example the Zariski topology cares more about closed sets).

Proof. Immediate from writing closed sets as complements of open sets, since the complement passes through the unions and intersections. □

3 September 3

3.1 Closure and limit points

Definition 3.1 (Interior and closure) — The **interior** $\mathring{A}$ of $A \subseteq X$ is the union of all open subsets of X contained in A. The **closer** $\overline{A}$ of $A \subseteq X$ is the intersection of all closed subsets of X containing A.

Proposition 3.2. The interior and closure are respectively maximal and minimal in the following sense:

(1) $\mathring{A}$ is the largest open set contained in A.

(2) $\overline{A}$ is the smallest closed set containing A.

This is of course immediate from the definition. Note also that $A = \mathring{A}$ is equivalent to A being open and $A = \overline{A}$ is equivalent to A being closed.

The closure of a set can also be defined in terms of more local properties of the set. To make this idea more precise, we define the following:

Definition 3.3 (Limit point) — Let $A \subseteq X$. Then $x \in X$ is a **limit point** of A if any open set containing x intersects A at a point besides x, i.e. $U \cap A$ contains a point not equal to x for all open $U \ni x$.

Theorem 3.4 (Limit points provide closure)**.** Given $A \subseteq X$, let A' be the set of limit points of A in X. Then $\overline{A} = A \cup A'$.

Proof. If $x \notin A \cup A'$, then there is a neighborhood $U \ni x$ that does not intersect A. This means that $X \setminus U$ is a closed set containing A that doesn't contain x, so x is also not in $\overline{A}$. Thus $\overline{A} \subseteq A \cup A'$.

Conversely, if $F \supseteq A$ is closed and x is not in F, then x is contained in the open set $X \setminus F$, which does not intersect A at all, so x isn't a limit point for A. Therefore any closed set containing A contains $A \cup A'$, so $\overline{A}$ contains $A \cup A'$.

We have shown that $\overline{A}$ and $A \cup A'$ contain each other, so the two sets are equal. □

At this point a useful condition to impose on topological spaces is that they be **Hausdorff**, meaning that given any distinct x and y in X, there exist open $U \ni x$ and $V \ni y$ that are disjoint. This ensures that our space has enough open sets to meaningfully separate points from each other; most topological spaces we encounter in the real world are Hausdorff.

Exercise 3.5. Show that $\mathbb{R}^n$ with the standard topology is Hausdorff.

Proposition 3.6. Finite subsets of Hausdorff spaces are closed.

Proof. It suffices to prove this for singleton sets, since finite subsets are finite unions of singleton sets. Indeed, if $x \in X$, then for any $y \in X$ not equal to x there are neighborhoods $U \ni x$ and $V \neq y$ such that $U \cap V = \varnothing$ so in particular $x \notin V$; this implies that any $y \neq x$ is not a limit point of $\{x\}$, so $\{x\}$ is closed in X. $\square$

3.2 Continuous functions

The standard definition of a continuous function between two metric spaces asks that for any threshold ε, there is a δ such that two points that are at most δ apart in the domain well be at most ε apart in the codomain. Said differently, given $x \in X$, the ball $B_Y(f(x), \varepsilon)$ has pre-image containing some open ball $B_X(x, \delta)$ in X. By taking sub-balls of $B_Y(f(x), \varepsilon)$ centered at each point in this ball, we find that this forces $f^{-1}(B_Y(f(x), \varepsilon))$ to be open in X since we have an open ball in X centered at each point of the pre-image while still living inside the pre-image. Because the balls in Y are a basis for Y's topology, this shows that the pre-image of any open set in Y must be open in X (since unions pass through pre-images).

What we have done, in short, is eliminated any reference to the metrics on X and Y in our definition of continuity, referring instead to only topological data! This is the insight behind the definition of continuous functions between topological spaces:

Definition 3.7 (Continuous function, homeomorphism) — A function $f : X \to Y$ between topological spaces is **continuous** if, for every open $V \subseteq Y$, the pre-image $f^{-1}(V) \subseteq X$ is open. If f is bijective and f^{-1} is continuous as well, then f is a **homeomorphism** and X and Y are **homeomorphic**.

Exercise 3.8. Exhibit a homeomorphism between $\mathbb{R}$ (with the standard topology) and any open interval in $\mathbb{R}$ (with the subspace topology from $\mathbb{R}$).

Exercise 3.9. Show that $f : [0, 2\pi) \to S^1 \subseteq \mathbb{R}^2$ (where S^1 is the unit circle in $\mathbb{R}^2$) defined as $f(t) = (\cos t, \sin t)$ is continuous and bijective but not a homeomorphism, because its inverse is not continuous.

4 September 8

4.1 Product topology

Recall that given finitely many topological spaces $X_1, X_2, \ldots, X_n$ we can impose a topology on $X_1 \times X_2 \times \cdots \times X_n$ by decreeing its basis to be

$$\{U_1 \times \cdots \times U_n : U_i \overset{\text{open}}{\subseteq} X_i\}.$$

If instead we are given an arbitrary collection $\{X_\alpha\}_{\alpha \in A}$ of topological spaces, then one way to impose a topology on $\prod_{\alpha \in A} X_\alpha$ is:

Definition 4.1 (Box topology) — The **box topology** on the product $\prod_{\alpha \in A} X_\alpha$ of topological space $\{X_\alpha\}_{\alpha \in A}$ is given by the basis

$$\left\{ \prod_{\alpha \in A} U_\alpha : U_\alpha \overset{\text{open}}{\subseteq} X_\alpha \right\}$$

We will shortly see that this topology on the product is insufficient, and we prefer the following one instead:

Definition 4.2 (Product topology) — The **product topology** on $\prod_{\alpha \in A} X_\alpha$ is given by the subbasis

$$\mathcal{S} = \bigcup_{\alpha \in A} \left\{ \pi_\alpha^{-1}(U_\alpha) : U_\alpha \overset{\text{open}}{\subseteq} X_\alpha \right\},$$

where π_α is the projection onto X_α.

Exercise 4.3. What is the basis of the product topology given by the subbasis $\mathcal{S}$? (i.e. what is the collection of finite intersections of sets in $\mathcal{S}$?)

Exercise 4.4. Show that the box and product topologies agree for products of finitely many spaces.

In other words, the product topology is defined as the coarsest topology that makes the π_α maps continuous. The π_α are continuous because we are specifically saying that the pre-image of any open set in U_α is open in the product topology — this is literally the definition of $\mathcal{S}$. It shortly follows that this is also the coarsest topology allowing the projections to be continuous.

Theorem 4.5 (Continuity via component functions)**.** Suppose $\prod_{\alpha \in A} X_\alpha$ is given the product topology. Then, a map $f\colon Y \to \prod_{\alpha \in A} X_\alpha$ is continuous if and only if the components functions $f_\alpha\colon Y \to X_\alpha$, defined as $\pi_\alpha \circ f$, are all continuous.

Proof. If f is continuous then the result is immediate because compositions of continuous functions are continuous. (This hasn't been proven in these notes yet; go check it.)

On the other hand, if the f_α are continuous, we can verify that f is continuous by checking that $f^{-1}(U)$ is open where U is an element of the subbasis $\mathcal{S}$ that generates the product topology. Taking $U = \pi_\alpha^{-1}(U_\alpha)$,

$$f^{-1}(U) = f^{-1}(\pi_\alpha^{-1}(U_\alpha)) = (\pi_\alpha \circ f)^{-1}(U_\alpha) = f_\alpha^{-1}(U_\alpha). \qquad \square$$

The kicker is that this theorem fails for the box topology!

Example 4.6 (Box topology is bad)
Let $\mathbb{R}^\omega$ be the product of countably many copies of $\mathbb{R}$, endowed with the box topology. Consider $f\colon \mathbb{R} \to \mathbb{R}^\omega$ given by $f(t) = (t, t, t, \dots)$; this definitely should be continuous in any reasonable topology. However,

$$U = \prod_{n=1}^{\infty} (-1/n, 1/n)$$

is an open set whose pre-image is $\bigcap_{n=1}^{\infty}(-1/n, 1/n) = \{0\}$, which isn't open in $\mathbb{R}$! (More abstractly, being able to take too many sets in the product defining U forced an infinite intersection of open sets in the pre-image, which needn't be open.)

4.2 Metric spaces

Recall the definition of a metric space:

Definition 4.7 (Metric space) — A **metric space** is a set X endowed with a **metric** $d\colon X \times X \to \mathbb{R}_{\geq 0}$ such that

(1) $d(x, y) = 0$ if and only if $x = y$;

(2) $d(x, y) = d(y, x)$ for all $x, y \in X$;

(3) $d(x, y) \leq d(x, y) + d(z, y)$ for all $x, y, z \in X$.

Example 4.8
$\mathbb{R}^n$ with the distance function

$$d(x, y) = \sqrt{\sum_{n=1}^{\infty} (x_i - y_i)^2}$$

is a metric space; the only nontrivial part of checking this is showing the triangle inequality. One way to do this is to anchor z to the origin (since $d(x-z, y-z) = d(x, y)$) and then expand and use the Cauchy-Schwarz inequality.

Then, metric spaces can be given a topology, as seen in analysis class:

Definition 4.9 (Metric topology) — The **metric topology** on metric space (X, d) has basis consisting of **metric balls**, i.e. the sets $B_\varepsilon(x) = \{y \in X : d(x, y) < \varepsilon\}$.

Exercise 4.10. Verify that this is a basis for a topology by showing that the intersection of two balls satisfies the egg yolk property (that is, the intersection of the balls must be open, hence a countable union of balls).

Exercise 4.11. Show that $\rho\colon \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}_{\geq 0}$ defined by $\rho(x, y) = \max_i |x_i - y_i|$ is a metric on $\mathbb{R}^n$ and that the corresponding topology is also the Euclidean topology. What do the metric balls in this space look like?

Theorem 4.12 (Equivalence of metrics). Let d and d' be two metrics on X. Then the topologies generated by d and d' are the same if and only if there exists a constant $C \geq 1$ such that

$$B^{d'}_{C^{-1}\varepsilon}(x) \subseteq B^{d}_{\varepsilon}(x) \subseteq B^{d'}_{C\varepsilon}(x)$$

for all $x \in X$ and $\varepsilon > 0$.

Definition 4.13 (Metrizable) — A topological space X with topology $\mathcal{T}$ is **metrizle** if there is a metric on X whose metric topology is $\mathcal{T}$.

We will talk about metrizability Much Later.

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Proposition 5.1. Metric spaces are Hausdorff.

Proof. Let $x \neq y$ be in X. Then, $d(x, y) = d_0 > 0$. Take the open balls $B_{d_0/2}(x) \ni x$ and $B_{d_0/2}(y) \ni y$. These two don't intersect, since if z were in their intersection, then by triangle inequality

$$d_0 = \frac{d_0}{2} + \frac{d_0}{2} > d(x, z) + d(z, y) > d(x, y) = d_0,$$

which is a contradiction. □

Theorem 5.2 (ε-δ continuity). A function $f\colon (X, d_X) \to (Y, d_Y)$ is continuous if and only if for each $x_1, x_2 \in X$ and $\varepsilon > 0$, there exists $\delta > 0$ (possibly depending on x_1, x_2) such that

$$d_X(x_1, x_2) < \delta \implies d_Y(f(x_1), f(x_2)) < \varepsilon.$$

Exercise 5.3. Show that the operations $+$, $-$, $\times$, $\div$ are continuous as maps from $\mathbb{R}^2$ to $\mathbb{R}$.

Definition 5.4 (Convergence) — Let $x_1, x_2, \ldots \in X$ be a sequence in topological space X. We say that $(x_n)_n$ **converges** to $x \in X$ if every open set containing x contains cofinitely many terms of the sequence.

Proposition 5.5. If X is Hausdorff and a sequence $(x_n)_{n\in\mathbb{N}}$ converges to both x and x' in X, then $x = x'$.

Proof. If $x \neq x'$ then we pick disjoint open sets containing x and x'; it is not possible for both of these open sets to have cofinitely many points of the sequence. □

Theorem 5.6 (Convergent sequences provide limit points)**.** Let $A \subseteq X$. If $(x_n)_{n\in\mathbb{N}}$ is a sequence of points in A with $x = \lim_{n\to\infty} x_n$, then $x \in \overline{A}$. If X is a metric space, then the converse is true; that is, if $x \in \overline{A}$, then there is a sequence of points in A converging to A.

Remark 5.6.1. More generally the converse holds in first countable spaces.

Proof. If $x \in A$ then we are done. Assuming $x \notin A$, the definition of convergence says that any neighborhood of x contains points in the sequence (x_n), which are in A; none of these x_n are equal to x by assumption, so x is a limit points for A, hence in $\overline{A}$.

For the converse, we again are immediately done if $x \in A$ because we can take the constant sequence. Otherwise, take successively smaller open balls $B_{1/n}(x)$ around x; since these are open and contain x, they intersect A, so take $x_n \in B_{1/n}(x) \cap A$. Since any open set containing x contains some ε-ball and for any $\varepsilon > 0$ there exists $n \in \mathbb{N}$ such that $\frac{1}{n} < \varepsilon$, we get that any open set containing x contains cofinitely many of the x_n. □

We can use sequences to phrase continuity if the spaces are nice enough, just like we did with maps between metric spaces:

Theorem 5.7 (Sequential continuity if domain is metric space)**.** Let X be a metric space and Y a map. Then f is continuous if and only if for every sequence $x_n \to x$ in X we have $f(x_n) \to f(x)$ in Y.

todo

Proof. □

Exercise 5.8. Show that $f\colon \mathbb{R}^2 \setminus \{(0,0)\} \to \mathbb{R}$ defined by $f(x,y) = \frac{xy}{x^2+y^2}$ cannot be extended to a continuous function on $\mathbb{R}^2$.

Definition 5.9 (Uniform convergence) — Let $f_n\colon X \to (Y, d_Y)$ be a sequence of functions (where X can be anything it wants to be). They **converge uniformly** to $f\colon X \to Y$ if given $\varepsilon > 0$ there exists N such that $d(f_n(x), f(x)) < \varepsilon$ for all $n \geq N$ and $x \in X$.

In other words, $f_n(x) \to f(x)$ at roughly the same rate for all $x \in X$.

Exercise 5.10. Let $f_n\colon [0, \frac{1}{2}] \to \mathbb{R}$ be defined by $f_n(x) = x^n$. Show that f_n converges to the zero function uniformly. What if the domain is instead $[0, 1]$ or $[0, 1)$?

6 September 15

6.1 Quotient topology and quotient maps

Another way to construct topological spaces is by gluing pieces of spaces together, such as rolling up a square into a cylinder and then rolling that cylinder into a donut, or simply rolling an interval into a circle. One can make this idea precise as follows:

Definition 6.1 (Quotient topology) — Let X be a topological space endowed with an equivalence relation $\sim$. Then, the set of equivalence classes $X/\sim$ is the **quotient space** of X modulo $\sim$, which is topologized by declaring $U \subseteq X/\sim$ to be open if and only if the union of the equivalence classes in U, i.e. $\bigcup_{[x]\in U}[x]$, is open in X. Equivalently, $U \subseteq X/\sim$ is open if and only if $p^{-1}(U) \subseteq X$ is open, where the projection $p\colon X \to X/\sim$ is the map $p(x) = [x]$.

Note of course that this means p is a continuous map $X \to X/\sim$.

Example 6.2 (S^1 is a quotient of $[0, 2\pi]$)

Give $[0, 2\pi]$ the equivalence relation whose only nontrivial relation is $0 \sim 2\pi$. Then $X = [0, 2\pi]/\sim$ is homeomorphic to the unit circle $S^1 = \{(\cos\theta, \sin\theta) \in \mathbb{R}^2 : \theta \in [0, 2\pi)\} \subseteq \mathbb{R}^2$ via the map $h\colon X \to S^1$ defined as $h(x) = (\cos x, \sin x)$.

Exercise 6.3. Check that h really is a homeomorphism $X \to S^1$.

Example 6.4

Let $X = \overline{B_1(0)} \subseteq \mathbb{R}^2$ be the closed disk of radius 1 centered at 0, and let $\sim$ be the relation such that any two points on the boundary S^1 are equivalent. This is actually homeomorphic to the sphere $S^2 \subseteq \mathbb{R}^3$! We can imagine the boundary collapsing onto the north pole, and then it is not too hard to convince yourself that open sets around the north pole in S^2 correspond to open sets in X.

Exercise 6.5. Let $X = [0, 1]^2$ and identify points with the relation $(x, 0) \sim (x, 1)$ for all $x \in [0, 1]$ and $(0, y) \sim (1, 0)$ for all $y \in [0, 1]$. What shape does $X/\sim$ look like? (Hint: identify the two pairs of edges one at a time.)

We can also get quotient maps in a sort of backwards way, by deriving the underlying equivalence relation via a map that behaves like the projection map.

Definition 6.6 (Quotient map) — Let $p\colon X \to Y$ be a surjective map such that $p^{-1}(U) \subseteq X$ is open if and only if $U \subseteq Y$ is open.

In this case Y can be thought of as the quotient of X by the relation $x \sim x'$ whenever $x, x' \in p^{-1}(Y)$ for some $y \in Y$.

Exercise 6.7. Show that the map $p\colon [0, 2\pi] \to S^1$ given by $p(t) = (\cos t, \sin t)$ is continuous and surjective but is not a quotient map.

Proposition 6.8. Let $X/{\sim}$ be a quotient space of X with projection map $p\colon X \to X/{\sim}$ and suppose that there is a continuous map $f\colon X \to Z$ that is constant on $p^{-1}(y)$ for each $y \in X/{\sim}$. Then, there exists a unique continuous map $\widetilde{f}\colon X/{\sim} \to Z$ such that $\widetilde{f} \circ p = f$.

check continuity

Proof. For a given $y \in X/{\sim}$, define $\widetilde{f}(y)$ to be the value of f at any point in $p^{-1}(y)$. It is easy to see that $\widetilde{f} \circ p = f$, so it remains to check continuity. ☐

Example 6.9 (Real projective space)

$\mathbb{R}P^n = S^n/{\sim}$ where $\sim$ identifies antipodal points. (TODO elaborate)

Given $f\colon S^1 \to \mathbb{R}$ defined by $f(x, y) = y^2$ if $y \geq 0$ and $f(x, y) = 0$ if $y < 0$, does there exist a corresponding $\widetilde{f}$?

6.2 Connectedness

Definition 6.10 (Connectedness) — Let X be a topological space with $X = U \cup V$ where U, V are nonempty disjoint open sets; we say that $\{U, V\}$ is a **separation** of X and that X is **connected** if it does not admit a separation.

examples

7 September 17

Proposition 7.1. X is connected if and only if the only subsets of X that are both open and closed are X and $\varnothing$.

Proof. There exists a set U that is both open and closed if and only if there is a separation of X (which is equivalent to X being disconnected), since $X = U \cup V$ is the union of two disjoint nonempty open sets if and only if $U = X \setminus V$ is open by definition and closed by being the complement of V. ☐

Theorem 7.2 (Unions of connected sets). Let $\{U_\alpha\}$ be a collection of connected sets in X. If the intersection of the U_α is nonempty, then $\bigcup_\alpha U_\alpha$ is connected.

lazy

Solution. Suppose for the sake of contradiction that $\bigcup_\alpha U_\alpha$ has a separation $V \sqcup V'$. ☐

Theorem 7.3 (f(connected) = connected)**.** Let $f\colon X \to Y$ be continuous. If X is connected then $f(X)$ is connected.

Proof. Suppose the subspace $f(X) \subseteq Y$ has a separation $U \sqcup V$. Then $U = \widetilde{U} \cap f(X)$ and $V = \widetilde{V} \cap f(X)$ for some $\widetilde{U}$, $\widetilde{V}$ open in Y. Then the pullbacks of $\widetilde{U}$ and $\widetilde{V}$ are clearly open (by continuity) and nonempty, disjoint, and cover X (by basic properties of pullbacks of sets), so $f^{-1}(\widetilde{U})$ and $f^{-1}(\widetilde{V})$ are a separation of X, hence X is disconnected if $f(X)$ is disconnected. □

Definition 7.4 (Path connected) — X is **path connected** if there exists a continuous path between any two points $x, y \in X$; that is, there is a continuous function $f\colon [0,1] \to X$ such that $f(0) = x$ and $f(1) = y$.

Proposition 7.5.

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comb space (line segments and $(0,1)$)

8 September 22

Lemma 8.1. Let X be a topological space and let $A \subseteq X$ be connected. If $A \subseteq B \subseteq \overline{A}$ then B is connected.

Corollary 8.2. The deleted comb space is connected.

Proof of Lemma 8.1. Assume B is disconnected, so there exist nonempty sets $U, V \subseteq B$ that are open in B whose union is B. Let $\widetilde{U} = U \cap A$ and $\widetilde{V} = V \cap A$. Then $\widetilde{U}$ and $\widetilde{V}$ are open in A and $A = \widetilde{U} \cup \widetilde{V}$, so it remains to show that $\widetilde{U}$, $\widetilde{V}$ are nonempty. By definition of the subspace topology there exists $\widehat{U}$ that is open in X such that $U = \widehat{U} \cap B$, hence $\widetilde{U} = \widehat{U} \cap A$. By the closure condition either some point of A is in $\widehat{U}$, in which case we're done, or $\widehat{U}$ contains a limit point of A, in which case it must still contain a point of A, so we are still done (because either way $\widetilde{U}$ is nonempty, hence so is $\widetilde{V}$ by symmetry). □

Theorem 8.3 (Intervals are connected)**.** Intervals and rays in $\mathbb{R}$ (i.e. sets of the form (a,b), $(a,b]$, $[a,b)$, or $[a,b]$ such that $-\infty \le a \le b \le \infty$, with strict inequalities if the corresponding endpoint is closed) are connected.

finish

Proof. Let I be an interval and $U \cup V$ is a separation. blah blah □

Corollary 8.4 (Intermediate value theorem)**.** blah blah

Corollary 8.5 (Brouwer fixed point theorem, 1D).

Exercise 8.6. Does the equation $x^3 + x = e^{\sin(x^3)} + 10x^2 - \log(|x| + 1)$ have a solution in $\mathbb{R}$?

9 September 24

Given a topological space, one way to look at the connectedness/path-connectedness properties can be understood by looking at maximal (path-)connected subsets.

Definition 9.1 ((Path) connected components) — A **connected component** (or just **component**) of a space X is a maximal connected subset of X. Equivalently, it is an equivalence class of X under the relation $a \sim b$ if there exists a connected subset of X containing both a and b. Analogously, **path components** are connected components but for path connectedness.

It is easy to show (and you should convince yourself if it isn't obvious) that any (path) connected subset of X can only lie in one component of X.

Definition 9.2 (Local (path) connectedness) — We say that X is locally (path) connected if for any $x \in X$ and neighborhood $U \ni x$ there exists an open (path) connected $V \subseteq U$ such that $V \ni x$.

Example 9.3

Local connectedness and connectedness do not imply each other.

- $\mathbb{R}$ (or any connected manifold) is locally connected and locally connected.
- $[0, 1) \cup (1, 2]$ is locally connected but clearly not connected.
- The comb space is connected but not locally connected (find a neighborhood that breaks!).
- $\mathbb{Q}$ is neither connected nor locally connected.

Theorem 9.4 (Locally connected means components of open are open). X is locally connected if and only if each component of any open $U \subseteq X$ is open.

Theorem 9.5 (Path component properties). (1) Path components lie inside connected components.

(2) If X is locally path connected then the components and path components of X are the same.